Compound gear trains

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Calculate the velocity ratio needed.

- VR = 1200 ÷ 25 = 48

1b) Calculate the number of teeth gear A must have.

- $VR_2 = 45 \div 15 = 3$, $VR_3 = 48 \div 12 = 4$
- 48 = (32 ÷ A) × 3 × 4, so 32 ÷ A = 4
- A = 8 teeth

2a) Calculate the velocity ratio.

- VR = (50 ÷ 20) × (90 ÷ 30) = 2.5 × 3 = 7.5

2b) Calculate the output shaft speed.

- 750 ÷ 7.5 = 100 revs min^{-1}

3a) State the type of gear system.

- Compound gear train

3b) Calculate the velocity ratio of the system.

- VR = 2250 ÷ 150 = 15

3c) Calculate the ratio of gear B to gear A.

- 60 ÷ 12 = 5

3d) Calculate the number of teeth on gear D.

- D ÷ C = 15 ÷ 5 = 3
- D = 3 × 45 = 135 teeth

3e) State one advantage of plastic gears over steel gears.

- Lighter, cheaper, quieter or self-lubricating (less friction)

4a) Design a simple gear train where the output turns 5 times slower than the input. Sketch it using the correct symbols.

- Driven gear has 5 times the teeth of the driver, e.g. 8 T driving 40 T

4b) Design a compound gear train where the output turns 20 times slower than the input. Sketch it using the correct symbols.

- Two stages that multiply to 20, e.g. 8 T → 40 T (5) and 10 T → 40 T (4): 5 × 4 = 20